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Iron nitrate and 4-OH-TEMPO-cocatalyzed aerobic oxidation of isochromans†

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An efficient aerobic oxidation of isochromans and their derivatives with $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ and 4-OH-TEMPO as catalysts at room temperature has been developed. This method provides a convenient approach for synthesizing biologically active isochromanone-containing compounds. Optically active products could be prepared from optically active isochromans without racemization. This catalytic protocol can also be extended to the oxidation of xanthene and phthalan in satisfactory yields.

Introduction

Isochromanone derived structures broadly exist in natural products (Fig. 1).¹ Oxygen is environmentally friendly and inexpensive, hence, the aerobic oxidation of the benzylic C–H bonds of isochromans has attracted much attention.² Direct aerobic oxidation of isochromans was reported; however, a high temperature of 80 or 140 °C was required (Scheme 1a).³ Recently, photoredox catalysts,⁴ electrochemical catalysts,⁵ acid catalysts,⁶ organocatalysts,⁷ nitroxyl catalysts,⁸ doped carbon materials,⁹ and transition metal catalysts¹⁰ have been employed in the aerobic oxidation of isochromans. Iron-based catalysis has received much attention because of its natural abundance, low cost and environmentally friendly nature (Scheme 1b):¹¹ Xiao and coworkers reported $\text{Fe}(\text{OTf})_2$ with a PyBisulidine ligand with low conversion with 1,1'-oxido-isochromans as the byproducts;^{11a} in 2017, Shen *et al.* successfully applied $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ with KPF_6 for the aerobic oxidation of isochromans in MeCN at 80 °C;^{11b} later, this transformation was also realized at 80 °C by Feng and He *via* the incorporation of dppf with disulfide;^{11c} herein, based on our early reports with $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ catalysis,^{11d,12} we wish to report our recent observation that with the help of 4-OH-TEMPO, the $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ catalyzed aerobic oxidation of isochromans could be realized in high yields under very mild conditions of rt (Scheme 1c).

Results and discussion

Our study commenced with the aerobic oxidation of isochroman 1a (Table 1). The reaction was carried out with 5 mol% $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ and 5 mol% 4-OH-TEMPO in MeCN at room temperature with an oxygen balloon. To our delight, the desired lactone was obtained in 100% yield within 24 h (entry 1). With the encouraging preliminary result, we screened the solvents including DCM, DCE, CHCl_3 , toluene, MTBE and MeOH (entries 2–7) first. MeCN gave the best result and MeOH totally shut down the reaction. A range of TEMPO derivatives were investigated (entries 8–13). Considering the cost, we chose 4-OH-TEMPO as the catalyst. Lowering the loading of catalysts led to a lower yield (entry 1 vs. entries 14 and 15). Control experiments showed that both iron and 4-OH-TEMPO were necessary (entries 16 and 17). It is noteworthy that no dimeric byproduct 2a' was observed.^{11a}

The scope of isochromans was subsequently evaluated under the optimal conditions (Scheme 2). Both electron-neutral and -rich substituents of isochromans displayed high

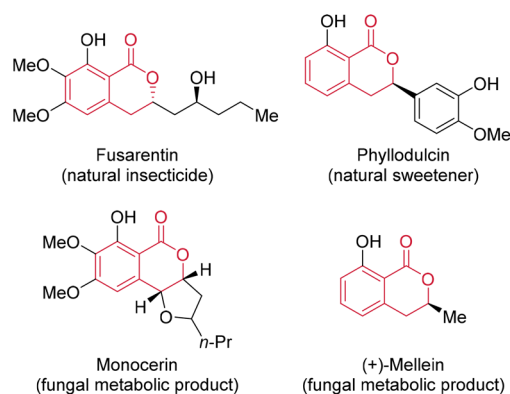


Fig. 1 Selected natural products containing isochromanone.

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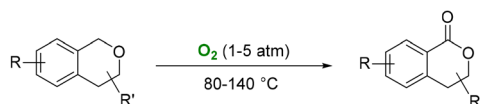
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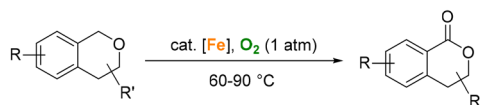
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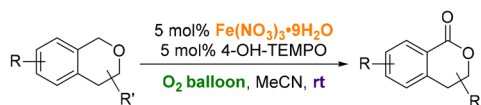
(a) Direct aerobic oxidation of isochromans



(b) Iron-catalyzed aerobic oxidation of isochromans



(c) This work



- ♦ Mild conditions ♦ Non-noble metal catalysis
- ♦ High yields ♦ Easy scale up

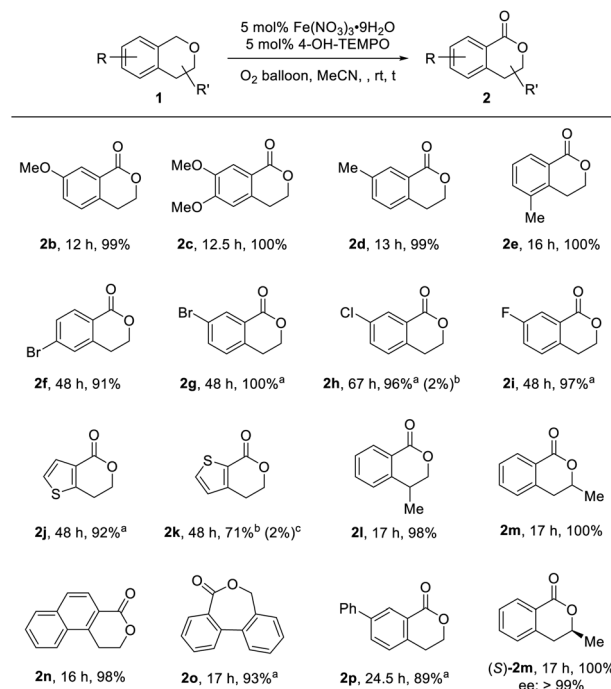
Scheme 1 Aerobic oxidation of isochromans.

Table 1 Optimization of the reaction conditions^a

Entry	X	Nitroxyl	Solvent	Recovery of 1a (%)	NMR yield of 2a (%)
1	5	4-OH-TEMPO	MeCN	0	100 (94 ^b)
2	5	4-OH-TEMPO	DCM	12	89
3	5	4-OH-TEMPO	DCE	16	84
4	5	4-OH-TEMPO	CHCl ₃	50	50
5	5	4-OH-TEMPO	Toluene	78	22
6	5	4-OH-TEMPO	MTBE	58	40
7	5	4-OH-TEMPO	MeOH	97	1
8	5	4-OMe-TEMPO	MeCN	0	100
9	5	4-OBz-TEMPO	MeCN	0	100
10	5	4-Oxo-TEMPO	MeCN	25	75
11	5	4-NHAc-TEMPO	MeCN	0	100
12	5	4-NH ₂ -TEMPO	MeCN	0	100
13	5	TEMPO	MeCN	1	99
14	2	4-OH-TEMPO	MeCN	15	85
15	1	4-OH-TEMPO	MeCN	50	42
16 ^c	5	4-OH-TEMPO	MeCN	100	0
17 ^d	5	—	MeCN	97	0

^a Reaction conditions: **1a** (1.0 mmol), Fe(NO₃)₃·9H₂O (*X* mol%) and 4-OH-TEMPO (*X* mol%) in solvent (1.25 M) with an oxygen balloon at room temperature and the recovery and NMR yield were determined by ¹H NMR analysis of the crude product. ^b Isolated yield. ^c Without Fe(NO₃)₃·9H₂O. ^d Without 4-OH-TEMPO.

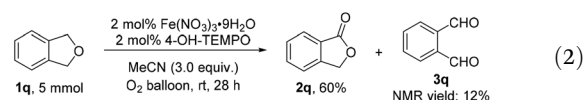
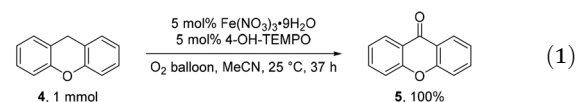
reactivity and provided lactones **2a–2e** in excellent yields. The reaction rate of electron-withdrawing substrates was relatively slow, thus, higher loadings of catalysts were required with excellent yields for products **2g–2i**. This transformation could

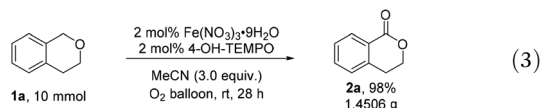


Scheme 2 Scope study. Reaction conditions: **1** (1.0 mmol), Fe(NO₃)₃·9H₂O (5 mol%) and 4-OH-TEMPO (5 mol%) in MeCN (0.8 mL) with an oxygen balloon at rt. All yields were isolated yields unless otherwise mentioned. Recovery was determined by ¹H NMR analysis of the crude reaction mixture using dibromomethane as the internal standard. ^a 10 mol% each of Fe(NO₃)₃·9H₂O and 4-OH-TEMPO were used. ^b Recovery of the starting material. ^c Isolated yield of thiophene-2,3-dicarbaldehyde **3k** is shown in parenthesis.

be extended to heteroaromatic derivatives **2j** and **2k**. Methyl-substituted products **2l** and **2m** may also be prepared in excellent yields. Naphthyl and biphenyl groups were also compatible (**2n–2p**). Furthermore, it is noted that chiral substrate (*S*)-**1m** (>99% ee) was oxidized under the standard conditions without racemization.

Next, we examined the scope of this method. In addition to isochromans, the reaction of xanthene **4** could be performed well under the standard conditions, giving xanthone **5** in 100% yield (eqn (1)). With 2 mol% each of Fe(NO₃)₃·9H₂O and 4-OH-TEMPO in MeCN, phthalan **1q** was successfully transformed into phthalide **2q** in 60% yield,^{3a,4b,c,f,n–p,6a,e,7,8l–n,10a–c,e,h,i,m,n,11c,13} although 12% yield of *o*-phthalaldehyde **3q** was observed (eqn (2)). In addition, the practicality has been demonstrated by applying 10 mmol of **1a**, affording 1.4506 g of **2a** in 98% yield (eqn (3)).





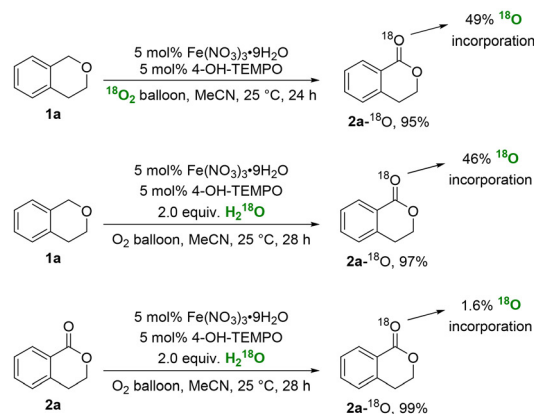
Several control experiments were carried out to gain insight into the reaction mechanism. Under an Ar atmosphere, 5 mol% each of two catalysts could also generate 13% yield of product **2a** (Table 2, entry 1). This discovery prompted us to increase the loading of each catalyst (Table 2, entries 2 and 3). As we expected, higher loadings of the catalysts led to higher yield of product **2a** even in the absence of oxygen. Thus, the presence of oxygen is critical for the catalytic cycle. Isotopic ^{18}O distribution experiments were also carried out using $^{18}\text{O}_2$ instead of $^{16}\text{O}_2$ or adding 2.0 equivalents of H_2^{18}O under the standard conditions to demonstrate the source of the oxygen atom in **2a** (Scheme 3A). It is clear that the oxygen atom on carbonyl comes from O_2 and/or H_2O . We also conducted control experiments to verify how water affects the reaction: with the addition of H_2O , the reaction slowed down (Table 3, entry 1 vs. entries 2 and 3). The reaction rate also reduced with the addition of 5 Å molecular sieves (Table 3, entries 4–6). Given that there is crystal water in the $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ catalyst and also water could be formed in the reaction process, additional water is unnecessary.

Based on the results of mechanistic studies and literature reports, a plausible mechanism is proposed (Scheme 3B). Nitrate is reduced to radical species, NO_2 and NO , by 4-OH-TEMPO under the promotion of Fe^{III} .^{12,14} NO_2 would abstract a benzylic hydrogen atom producing benzylic radical **1a'**. Next, **1a'** may couple with NO_2 affording nitrite ester **Int 1**, which would undergo HNO elimination to afford the target product **2a**.^{6b} Alternatively, the hydrolysis of **Int 1** may afford the hemiacetal **Int 2**, which could be further oxidized to give **2a**. The hemiacetal **Int 2** may also be generated from **Int 3**, which was formed by hydride transfer from the benzylic carbon of **1a** to the oxygen of the oxoammonium salt 4-OH-TEMPO⁺.¹⁵ Oxygen plays an important role in regenerating NO_2 by oxidizing NO .¹² The presence of peroxide intermediates cannot be excluded: radical **1a'** could react with O_2 to

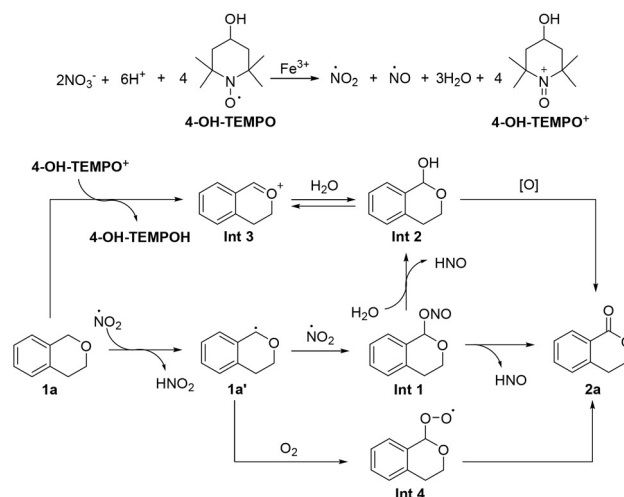
Table 2 Control experiments^a

Entry	X mol%	Recovery of 1a (%)	NMR yield of 2a (%)
1	5	85	13
2	15	62	35
3	40	14	86

^a Reaction conditions: **1a** (1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (X mol%) and 4-OH-TEMPO (X mol%) in MeCN (0.8 mL) at 25 °C. Recovery and NMR yield were determined by ^1H NMR analysis of the crude product. The reaction was carried out under an argon atmosphere and MeCN was degassed.

A) Isotopic ^{18}O Distribution Experiments

B) Proposed Mechanisms



Scheme 3 Mechanistic studies and proposed mechanisms.

Table 3 The effect of H_2O in aerobic oxidation of isochroman **1a**^a

Entry	X (equiv.)	Y (mg)	Recovery of 1a (%)	NMR yield of 2a (%)
1	0	0	0	100
2	1	0	44	53
3	2	0	62	35
4	0	20	7	93
5	0	40	15	85
6	0	60	65	25

^a Reaction conditions: **1a** (1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (5 mol%), 4-OH-TEMPO (5 mol%) and H_2O (X equiv.) or 5 Å molecular sieves (Y mg) in MeCN (0.8 mL) with an oxygen balloon at 25 °C. Recovery and NMR yield were determined by ^1H NMR analysis of the crude product.

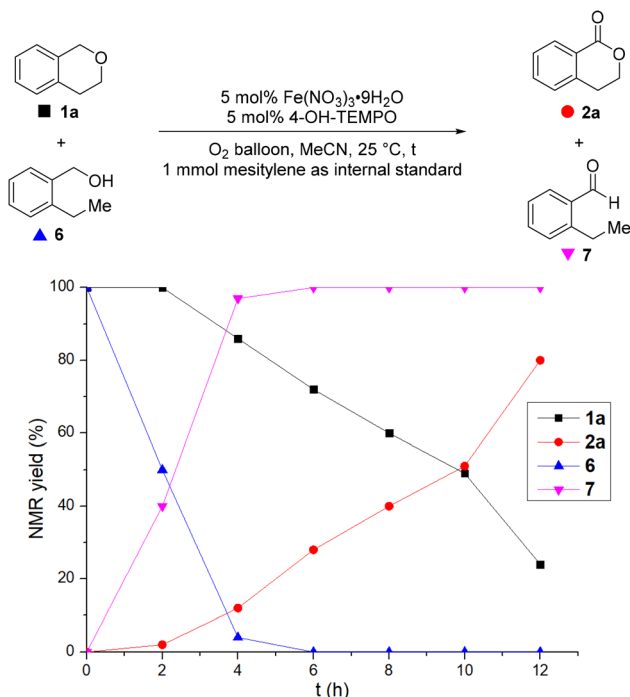


Fig. 2 The monitoring of iron nitrate and 4-OH-TEMPO-cocatalyzed aerobic oxidation of isochroman **1a** and *o*-ethylbenzyl alcohol **6**.

afford peroxy radical species **Int 4**, which could be further transformed to afford lactone **2a**.^{6b,7}

We also compared the efficiency of oxidation of the benzylic position by monitoring the iron nitrate and 4-OH-TEMPO-cocatalyzed aerobic oxidation of isochroman **1a** and *o*-ethylbenzyl alcohol **6** showing that the oxidation of *o*-ethylbenzyl alcohol **6** to aldehyde **7** is more efficient (Fig. 2).

Conclusions

In summary, Fe(NO₃)₃·9H₂O and 4-OH-TEMPO were found to be effective for the aerobic oxidation of the benzylic sp³ C-H bonds of isochromans under very mild conditions. This catalytic protocol can be extended to the oxidation of xanthene and phthalan in satisfactory yields as well. Based on mechanistic studies, a rationale has been proposed. Further investigations in this area are ongoing in our laboratory.

Conflicts of interest

There are no conflicts to declare.

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